What's new

Climbing the cosmic distance ladder: another sample chapter

Terence Tao · 21 November 2025

https://terrytao.wordpress.com/2025/11/21/climbing-the-cosmic-distance-ladder-another-sample-chapter/

tao2typ · converted to Typst 2026-10-14 (UTC)

Five years ago, I announced a popular science book project with Tanya Klowden on the cosmic distance ladder, in which we released a sample draft chapter of the book, covering the "fourth rung" of the ladder, which for us meant the distances to the planets. In the intervening time, a number of unexpected events have slowed down this project significantly; but I am happy to announce that we have completed a second draft chapter, this time on the "seventh rung" of measuring distances across the Milky Way, which required the maturation of the technologies of photography and spectroscopy, as well as the dawn of the era of "big data" in the early twentieth century, as exemplified for instance by the "Harvard computers".

We welcome feedback of course, and are continuing to work to complete the book despite the various delays. In the mean time, you can check out our instagram account for the project, or the pair of videos that Grant Sanderson (3blue1brown) produced with us on this topic, which I previously blogged about here.

Thanks to Clio Cresswell, Riley Tao, and Noah Klowden for comments and corrections.

Comments (16)

 **David Roberts** · 21 November 2025

Arg, the fonts and layout don't match! :-)

 **Terence Tao** · 21 November 2025

Can you clarify what the issue is? Of course the files will eventually be edited by the publisher, but it will be good to know what the problems are in advance.

 **David Roberts** · 21 November 2025

I just mean the two sample chapter pdfs, and it's more a reflection on me. Five years between two drafts can definitely see a shift in preferred style for a working document.

 **Anonymous** · 21 November 2025

I will definitely buy the book.

 **Anonymous** · 21 November 2025

The last line of page 24 has a spurious period between "system" and "to".
[Thanks, this will be corrected in the published version -T.]

 **Anonymous** · 21 November 2025

Yeay! I can finally comment on something in Terry Tao's blog: "riches nearly beyond imaging" on pg 28– maybe you meant "imagining"?
[Thanks, this will be corrected in the published version – T.]

 **Anonymous** · 22 November 2025

Dear Prof. Tao, Here are some typos I have collected. I will post more comments as I find them.
1) P88:
(a)"..were able today use..." should be " were able to use".
(b)"as a stars" should be " as stars"
2) P89:
(a)"rare occurance" should be "rare occurrence".
(b) "obseration" should be "observation".
3) P90:
(a)"millenia" should be "millennia".
I apologize for raising a topic unrelated to this post, but I want to ask you about: In your previous commons, you mentioned that there a couple of perspectives of modern maths, and you explained the set-theoretic perspective v. s. the categor- theoretic perspective——Could you also explained what the other perspectives (such as information-theoretic perspective, etc.) are? Also, when is it appropriate to adopt these perspectives?
[Thanks, this will be corrected in the published version. As for your other question, I refer you to https://mathoverflow.net/questions/38639/thinking-and-explaining -T.]

 **Anonymous** · 23 November 2025

P. 28, l. -3: The word "Lutheran" goes into the text margin.
[Thanks, this will be corrected in the published version – T.]

 **Anonymous** · 23 November 2025

P. 30, l. 12: The two hyphens should be m-dashes and the spaces around them should be deleted.

P. 31, l. 17: The hyphen should be an m-dash and the space around it should be deleted.
(I'm out of time for now but you should probably do a search and replace to check if this needs changing in other places.)
[Thanks, this will be corrected in the published version – T.]

Anonymous · 24 November 2025

> In a very slow process lasting millions of years, a typical star gradually sheds their mass through the radiation they emit, and steadily move down the main sequence until they run out of their primary nuclear fuel of hydrogen, at which point they leave the main sequence

should be either singular "it", or plural "stars"
[Corrected, thanks -T.]

Anonymous · 28 November 2025

On page 114, "Karl Scharzschild" should be replaced by "Karl Schwarzschild", the famous creator of the Schwarzschild solution to Einstein's field equations.
In the maths part you may want to consider typesetting physical units with the siunitx LaTeX package (such that units do not appear in maths font italics).
The chapter was very entertaining and informative to read!
[Thanks, this will be corrected in the final version of the ms. -T]

lebesguemeasure · 28 November 2025

On page 122, you say that, "the orbital plans of the stars will also tend to become aligned." I assume this should be planes?
-Aaron Cofer
[Thanks, this will be corrected in the final version of the ms. -T]

universallycollection781395846a · 29 November 2025

One small typo following eq. (7.4) in chapter 7.1.2: "*that relates the [[apparent]]* ***absolute*** *magnitude m0, the [[absolute]]* ***apparent*** *magnitude m, and the distance r in parsecs*"
"

[Thanks, this will be corrected in the final version of the ms. -T]

Anonymous · 8 December 2025

This looks very interesting professor!
Something which caught my eye in the sample draft for the "fourth rung" (perhaps already commented on elsewhere) is that the first sentence starts with "it's". However, elsewhere in the text you write "it is". This is a little bit inconsistent, and I would suggest consistently writing "it is".
-Clovis V

Anonymous · 17 April 2026

do you know how the ancient civilixations were able to distinguish time within two minutes using planetary positions? The D60 chart in vedic astrology is able to distinguish within two minutes using planetary positions (the chart shows difference of planet positions every two min or less and there are many charts d1 through d60 with different precisions (not every di occurs in charts)).



Terence Tao · 18 April 2026

My understanding is that many ancient cultures, including the Greeks, Indians, and Arabs, were able to draw upon a large amount of historical observational data dating back to the Babylonians

to help construct reasonably accurate planetary models. By the 5th century CE, the Indians also had managed to develop early versions of modern trigonometry, which allowed for somewhat simpler ways to adjust their models for what we now know to be heliocentric and Keplerian corrections, as compared with the corresponding corrections made (e.g., by al-Tusi) to the Ptolemaic model.